

CAPSTONE DELIVERABLES

Five artifacts that demonstrate enterprise architecture completeness - HFC Holistic Fitness Club

1

Architecture Diagram

Layers, components, trust boundaries and integrations

2

Agent Workflow Design

Roles, states, tools, handoffs, approvals and failure paths

3

Deployment Strategy

Runtime, scaling, resilience, environments and release

4

Security Model

Identity, authorization, secrets, privacy, guardrails and audit

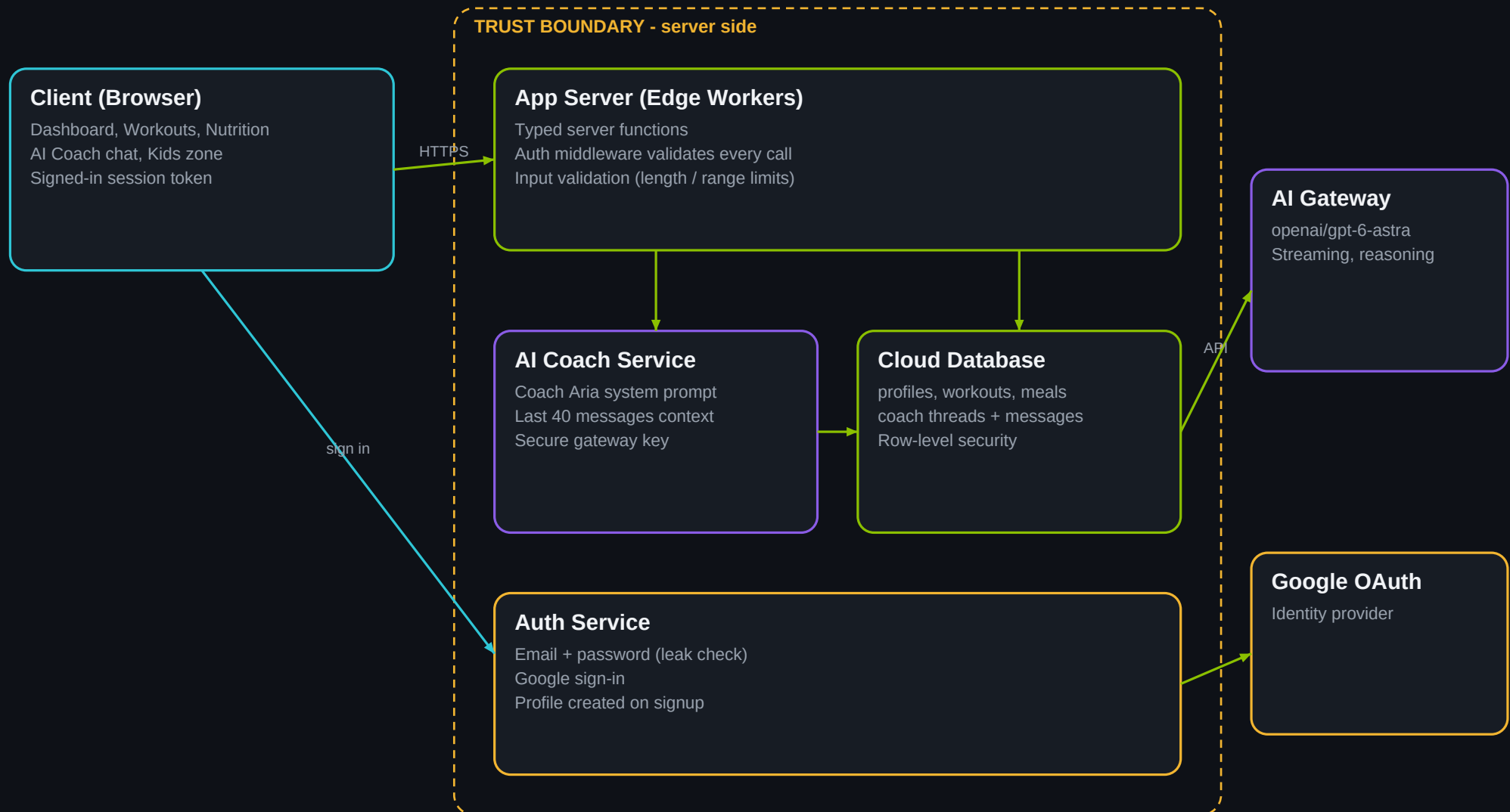
5

Monitoring Dashboard Design

Health, trace, quality, safety, cost and business outcomes

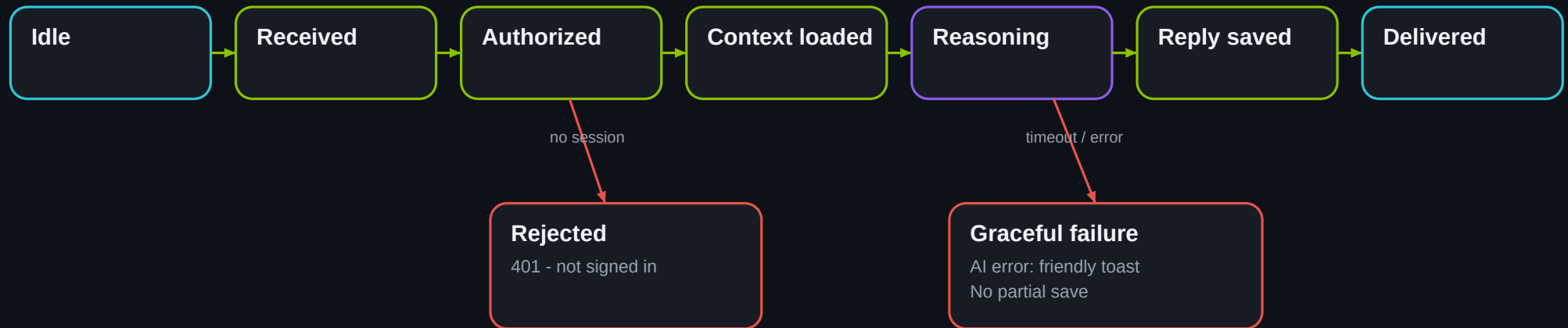
1. Architecture Diagram

Layers, components, trust boundaries and integrations



2. Agent Workflow Design

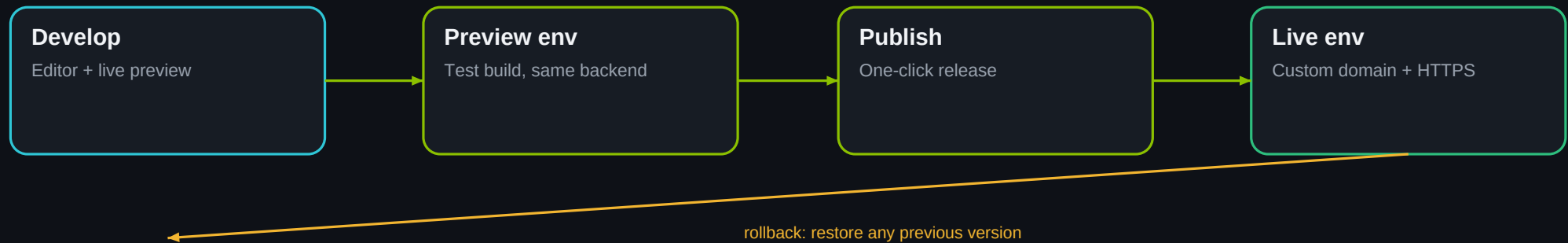
Roles, states, tools, handoffs, approvals and failure paths



Element	Design
Role	Coach Aria - concise, evidence-based fitness and nutrition coach (under 150 words)
Tools / context	Member's conversation history; own workouts and meals only
Handoff	Medical or injury questions redirected to a qualified professional
Approvals	Data changes (logging workouts / meals) require an explicit member action
Failure paths	Auth fail -> sign-in; AI fail -> retry prompt; DB fail -> error logged, nothing half-saved

3. Deployment Strategy

Runtime, scaling, resilience, environments and release

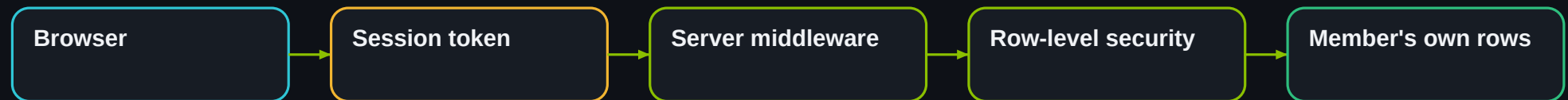


Concern	Strategy
Runtime	Global edge workers; server-side rendering; static assets on CDN
Scaling	Stateless servers auto-scale with traffic; managed database connection pooling
Resilience	Managed backups; bounded AI retries on 429/5xx; error boundaries in UI
Environments	Preview (testing) and Live (members); secrets stored per project, never in code
Release	Version history, instant rollback, build checks before publish

4. Security Model

Identity, authorization, secrets, privacy, guardrails and audit

Pillar	Control in HFC
Identity	Email/password with leaked-password check; Google OAuth; sessions with short-lived tokens
Authorization	Row-level security: <code>auth.uid() = user_id</code> on every table; server middleware re-checks
Secrets	AI and service keys live only on the server; browser holds publishable key only
Privacy	Workouts, meals and chats private per member; no cross-member reads
Guardrails	Input validation (zod); coach never gives medical diagnoses; length limits
Audit	<code>created_at</code> timestamps and owner IDs on all rows; server and AI request logs
Roles	Admin roles in a separate table checked by a secure function (never client-side)



Defense in depth: every request is checked at the server and again by the database.

5. Monitoring Dashboard Design

Health, trace, quality, safety, cost and business outcomes

Health

Uptime 99.9%
error rate < 1%

Trace

p95 latency
per request run ID

Quality

Reply rating
thumbs up / down

Safety

Flagged replies
medical redirects

Cost

AI requests / day
credits used

Business

Weekly active
streaks, retention

Sample: coach requests per day (illustrative targets)

